

PH-MAC: A Periodically Hybrid MAC Protocol for Wireless Sensor Networks

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Abstract

This paper proposes a novel protocol termed periodically hybrid MAC protocol (PH-MAC) for wireless sensor networks. In order to save energy and maximize the throughput we introduce the channel access mechanism TDMA into the PH-MAC. PH-MAC divides the whole transmission process into various PH-MAC periods, and each PH-MAC period is further divided into synchronization period, allocation period and TDMA period. The synchronization period is used to synchronize the neighboring nodes to prevent long-time clock drift; the allocation period is used to allocate timeslots to every node for communication; and the TDMA period is used for data communication. We present the designing techniques of PH-MAC, and compare the performance of PH-MAC with the simplified 802.11 DCF (no duty cycle), S-MAC (fixed duty cycle) and T-MAC (dynamic duty cycle) through simulation. The simulation results show that, PH-MAC not only improve the throughput of the network, but also save nearly half of the energy that T-MAC usually consumes.

Keywords: Sensor networks, MAC, TDMA, PH-MAC.

1 Introduction

For wireless sensor networks, energy is usually wasted in the processes of collision, overhearing, control packet overhead and idle listening. In this paper, we proposed a novel protocol termed periodically hybrid MAC protocol (PH-MAC). PH-MAC is designed to minimize the waste of energy in all the above processes. Besides energy efficiency, through PH-MAC, we are able to guarantee throughput and bandwidth utilization by allocating timeslots effectively in the contention period. PH-MAC proposes a new architecture of MAC protocol for data communication. PH-MAC is contention-based at allocation period and TDMA-based at the TDMA period. It uses contention period to allocate the timeslot to every node that needs to transmit data, and the TDMA part to transmit actual data. For every node at TDMA period, it just needs

to transmit data or receive data when the incoming data coming or outgoing data outgoing. What we want to do is to get the boring contention work done once and forever, though we could not keep it forever, just a moment (period) is enough for us. As we also need to deal with network changes and clock drift, so the PH-MAC runs periodically to adjust to those changes online. Under this architecture we get rid of contention, overhearing, idle listening and control packet overhead at the TDMA part, which could save much energy for the network.

Clearly, the energy used in the contention part is all wasted. So prolonging the TDMA part will save more energy, while the whole period will be prolonged, this will reduce the latency of the network. So we need to tradeoff latency with energy consumption.

The current MAC [1-4], [9], [10], which is designed for wireless sensor networks, can be broadly divided into contention-based and slotted protocol [4], [8]. Though a contention-based protocol is promising in terms of its simplicity, flexibility and robustness, It will suffer from contention, idle listening, overhearing, and this will waste lots of energy; while a slotted based protocol have no collisions, it requires an efficient time schedule, clock synchronization and adjusting to network topology changes.

The standardized IEEE 802.11 distributed coordination function (DCF) [1] is an example of the contention-based protocol, and is mainly built on the research protocol MACAW [2]. It is widely used in ad hoc wireless networks because of its simplicity and robustness to the hidden terminal problem. However the energy consumption in the wireless sensor network used this MAC is very high.

In order to reduce the waste of energy from collision, overhearing, idle listening and control packet overhead, S-MAC [3] induces a duty cycle into the 802.11, and use virtual clustering, message passing, collision avoidance, overhearing avoidance to save more energy. As this MAC protocol must predetermine the length of active part, which could not adjust to the network with variable load. So T-MAC [4] was proposed to dynamically adjust to the

variable load by inducing the TA threshold, which determines the minimal amount of idle listening per frame. This could minimize the idle listening and also dynamically adjust to the variable load.

We present PH-MAC, a periodically hybrid energy efficient MAC protocol for wireless sensor networks that minimizes energy waste. To demonstrate the effectiveness and measure the performance of our new MAC protocol, we have compared PH-MAC with other three existing MAC protocols (simplified 802.11 DCF, S-MAC and T-MAC) on network throughput, energy consumption and latency. Three simulators are designed for each of these protocols using C++.

2 Design of PH-MAC

Our MAC protocol is designed to reduce energy consumption and adjust to network changes and variable network load. This protocol is designed to reduce all energy consumption from all the sources that we have mentioned, i.e., idle listening, collision, overhearing and control overhead. In order to achieve the design goal, we have developed the MAC that consists of three major periods: synchronization period, allocation period and TDMA period. Before describing them we need to discuss our assumptions about the wireless sensor network and its applications.

2.1 Network Assumptions

Sensor networks are consisted of many small sensors and deployed in an ad hoc fashion. We expect all the sensors are the same, which means that all sensors have equal bandwidth, process capability and energy.

All sensor networks take advantage of short-range multi-hop communication instead of long-range communication to conserve energy [5]. Most communication will be between nodes as peers, rather than to a single base-station or sink node. So routing algorithm must be used to collect information in the network layer, i.e., directed diffusion [11], LEACH [12] and GAF [6]. As some sensors will fail when it is working, the routing algorithm must be robustness to the failure. Some sensors will move slowing along with the environment, i.e., sensors deployed on the sea, the routing algorithm could self-adjusting to the network topology changes periodically by inducing the periodical MAC protocol. We expect that the routing algorithm at the network layer will construct a routing tree to collect data. This routing tree will let every node knows its parent node which the data must be sent to and its child nodes which the data must be received from. This is what we need in the MAC design.

2.2 The Scheme of PH-MAC

The basic scheme of PH-MAC is shown in Figure 1, in this figure, PH-MAC is divided into three periods, they are synchronization period, allocation period, and TDMA period.

In section III-A, we have assumed every node will know its parent node and its child nodes once the routing tree is constructed. In the synchronization period, all nodes in the network maintain the same schedule by communicating with SYNC packets; in the allocation period, each node requests timeslots from his parent node for sending data, and allocate timeslots to his child node for receiving data; in the TDMA period, every node transmit packets in the timeslots that its parent node allocate to it, and receive data from its child nodes in the timeslots that it allocates to its child nodes.

2.3 Synchronization Period

In the synchronization period, the main task of the network is to synchronize the clock of all active nodes in the network, so as to minimize the effect of clock drift.

Clock synchronization is referred to virtual clustering, which is proposed by the authors of the S-MAC protocol [3]. While in our protocol, only the sink node who is the root node in the routing tree can start to send the SYNC packet at the start of one period, the other node in the network hears SYNC from another node, then follows the schedule in the SYNC packet and transmits its own SYNC packet. This will let all nodes in the network follow the sink's schedule. Any node that has not heard a SYNC packet must not transmit any data in this period, and this node will wait for the next synchronization period to receive the SYNC packet.

2.4 Allocation Period

In allocation period, the main work of the network is to allocate timeslots to every node who need to transmit data. In this period all nodes must contention for the channel to transmit packets, whereas there are two differences compared to 802.11 DCF:

- 1) The payload of allocation packet is not the data that the node needs to transmit, but the bitmap of the node's TDMA part;
- 2) All the packets follow the sequence of RTS/CTS/RTA/ALC/NOT, not the sequence of RTS/CTS/DATA/ACK as in the 802.11 DCF.

Every node has a bitmap, which will be used to describe local channel usage of TDMA period, every bit in the bitmap was mapped to a timeslot of TDMA period, that is to say, the first bit in the bitmap means whether the first timeslot in the TDMA period has been used by any node include himself. So at begin of every period, all the

timeslots are not used by any node, all bits in every node's bitmap should be zero.

Figure 2 shows the time schedule of RTS/CTS/RTA/ALC/NOT which is used in the allocation part to allocate timeslot.

We adopt the RTS/CTS mechanism to address the hidden terminal problem [3], what's more, by inducing network allocation vector (NAV) [1] we can reserve the channel to transmit the incoming allocation packets, e.g. RTA, ALC, NOT. After occupying the channel by using RTS/CTS mechanism, the node that has packets to transmit, needs to load his current bitmap state and the number of needed timeslots into the payload of the request to allocate (RTA) packet, and send it to the destination node. When the destination node receives the RTA packet, it gets the bitmap of the source node, by Boolean or the source node's bitmap and the destination node's bitmap together, the destination node get the bitmap of all available timeslots, then the destination node can allocate some of the available timeslots to the source node, and load the corresponding bitmap and the number of allocated timeslots to the allocation (ALC) packet. After received the ALC packet, the source node must first update its bitmap by Boolean or its bitmap with the bitmap in the ALC packet and then load the bitmap into the notice (NOT) packet, and broadcast the NOT packet to let its neighbors know that it has occupied those timeslots. If any neighbor of source and destination nodes has received RTA, ALC or NOT packet which is not routing to himself, it must update its bit-map by Boolean or its bitmap with the bitmap in the packet. The above procedure is given in Algorithm 1 as a pseudocode.

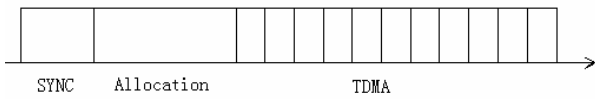


Figure 1 Scheme of PH-MAC: SYNC period, Allocation period, and TDMA period

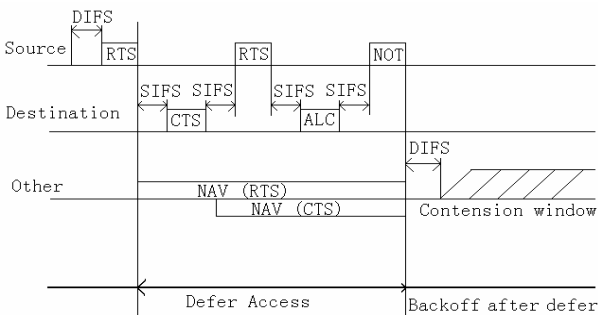


Figure 2 RTS/CTS/TC/ALC/NOT and NAV setting of the allocation period of PH-MAC

Algorithm 1 Distributed allocation algorithm used in the allocation period. Every node in the network will maintain a bitmap and node state.

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1.  If the channel is idle, then
2.    If node have packets to send and have not got
      any timeslot allocated then
3.      If node state is 0
4.        Send out a RTS packet
5.        node state  $\leftarrow$  1, and set a timeout timer;
6.      end if
7.    end if
8.    If timeout occurred then
9.      node state  $\leftarrow$  0
10.   end if
11.  If the channel is not idle, then
12.    If node get a packet suffered collision then
13.      Discard this packet.
14.    end if
15.    If node get a packet correctly then
16.      if packet is not addressed to this node then
17.        if this packet is RTS or CTS then
18.          Update NAV to reserve channel.
19.        end if
20.        if this packet is RTA or ALC or NOT then
21.          Update the node's bitmap by Boolean
            or the bitmap in the packet and the
            node's bitmap.
22.        end if
23.        if packet is addressed to this node then
24.          if this packet is RTS and node state = 0 then
25.            Send out CTS to reply
26.            node state = 2, and set a timeout
              timer.
27.          end if
28.          if this packet is CLS and node state = 1 then
29.            Load the bitmap and the number of
              needed timeslots into the RTA packet,
              and send it.
30.            node state  $\leftarrow$  3, and set a timeout
              timer.
31.          end if
32.          If packet is RTA packet and node state=2 then
33.            Boolean or the bitmap in the RTA packet
              and the node's bitmap, this node will get
              the available bitmap could be used to
              allocate.
34.            According to the available bitmap, this
              node will try to allocate the needed
              timeslots to the source node.
35.            Load the allocated timeslots to ALC
              packet
36.            Send out the ALC packet
37.            node state  $\leftarrow$  0, and set a timeout timer.
38.          end if
39.          If the packet is ALC packet and node state=
              3 then
40.            Load the allocated bitmap in the ALC packet
              into the NOT packet, and send it out.
41.            node state  $\leftarrow$  0;
42.          end if
43.        end if
44.      end if

```

One example of such schedule is shown in Figure 3.

In Figure 3, (a) shows all nodes' initial bitmap; (b) shows the source node are sending a RTA packet to the

destination node to ask for timeslots; (c) shows the all nodes' bitmap and the destination node's action after the RTA packet's arrival; (d) shows the destination node decide to allocate slot 1,2 and send out ALC packet with slot 1,2 to the source node; (e) shows all nodes' bitmap after ALC packet's arrival; (f) shows the source broadcast the NOT packet with its bitmap to all its neighbors; (g) shows all node's final state.

Loss of some packets in the sequence of RTS/CTS/RTA/ALC/NOT will have impact on packet transmissions in TDMA part. It does not matter if the RTS, CTS, or RTA packets have been lost, as the destination has not allocate any timeslots to this source node, so the source node can reconnect to the destination to ask for the timeslots. While loss of ALC will waste some of timeslots in the destination, as the destination together with its neighbors think that some one will use those timeslots, but the source node don't know those timeslots allocated to him, so those timeslots will be wasted. Loss of NOT will cause collision in the TDMA part, as the source node has occupied those timeslots, but the neighbors of the source exclude the neighbors of the destination node, don't know that some one has occupied those timeslots, this may cause collision in the TDMA part. In one word, loss of ALC and NOT will reduce the utilization of the TDMA part.

2.5 TDMA Period

In the TDMA period, every node sends packets in the timeslots allocated to him, and receive packets in the

timeslots that he allocates. Every data packet will be followed by an ACK packet if the destination node has received the packet.

In order to save consumed energy in the allocation period, we need to minimize the size of bitmap to reduce the time used in the contention part. So every timeslot can be used to transmit more than one packet. This will need fewer bits to map the timeslot of the TDMA period. The number of packets that can send in a timeslot could be set according to the application of the network and the data transmission rate of the RF model. But this will waste some time resource of the TDMA period and reduce the throughput of network. So we are making a tradeoff between the energy and the throughput. Figure 4 shows the schedule of the TDMA period. In the figure, every timeslot could be used to send two packets. In the first timeslot, node B send two packets to node A, then node C send one packet to node A in the next slot, finally, node D send one packet to node C in the third timeslot.

3 Simulation Results

In our simulations, we compare four protocols: simplified IEEE 802.11 DCF, S-MAC, T-MAC and PH-MAC. The purpose is to demonstrate the effectiveness of our protocol and to analyze how PH-MAC saves more energy than other protocols.

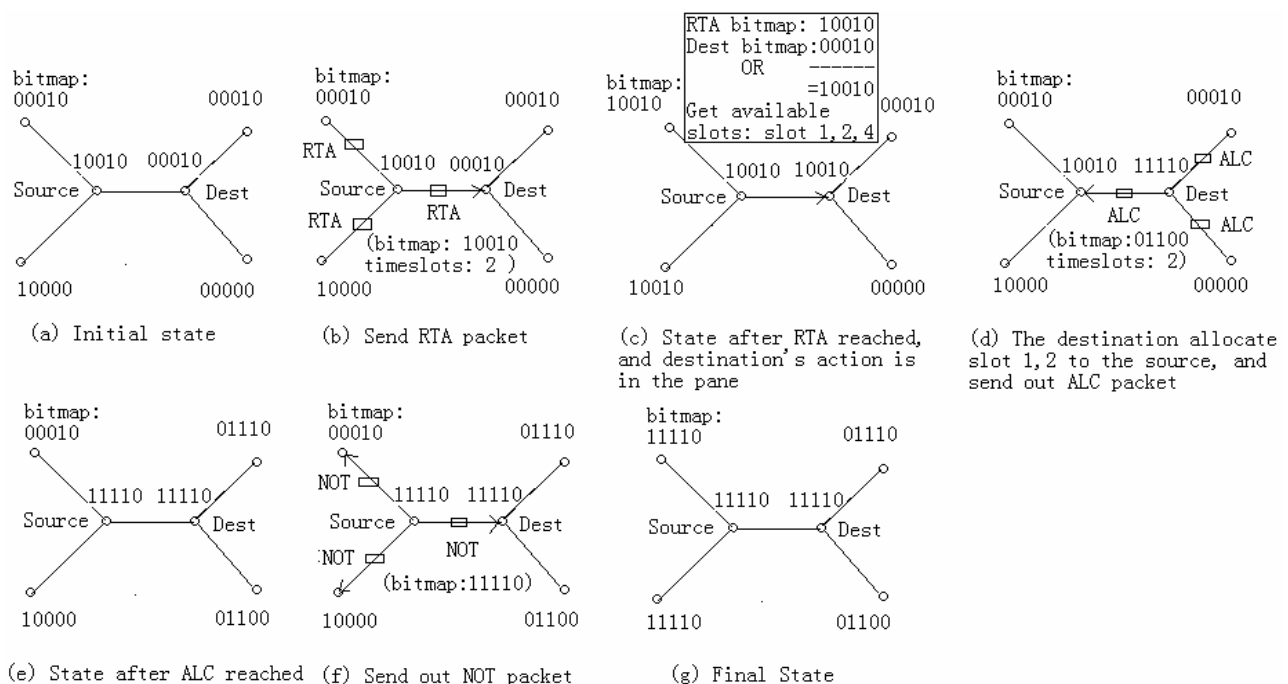


Figure 3 One example of the schedule of the allocation period

Table 1 Implementation detail and parameter setting in the simulator

PHYsical layer	
effective bit rate	46 kbps
DIFS	50 μ s
SIFS	10 μ s
802.11 [extends PHY]	
control packets	8 bytes
DATA packets	8 bytes and 32 bytes payload
CWmin	32
Cwmax	1024
S-MAC [extends 802.11]	
SYNC packets	10 bytes
length of one period	1 s
active period	300 μ s
T-MAC [extends S-MAC]	
activity time-out	15 ms
CWmin	32
CWmax	1024
P-MAC	
SYNC packets	10 bytes
control packets	8 bytes
allocation packets	10 bytes
DATA packets	8 bytes header and 32 bytes payload
CWmin	32
CWmax	1024
SYNC period	50 ms
Allocation period	150 ms
TDMA period	800 ms

3.1 Simulation Setup and Parameters Setting

The simulators of the four MAC protocols written in C++ are developed to evaluate their performance for a given wireless sensor network. We abstract the radio transceiver RF 1000 [7], and embed it into our simulator. The radio transceiver in our simulator provides a transmission rate of 46 Kbps, it has three working modes,

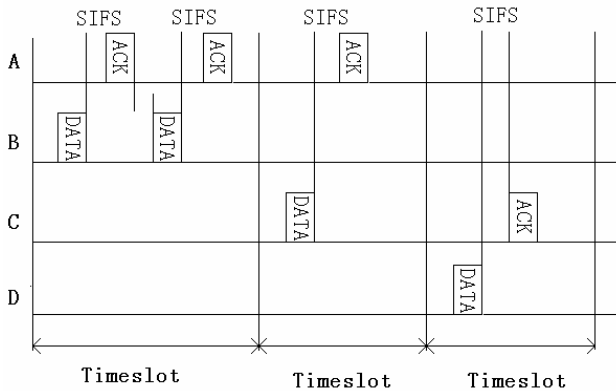


Figure 4 The schedule of the TDMA period, two packets. can be sent in one timeslot

i.e., the receiving mode, the transmitting mode and the sleeping mode, each drawing the input current of 4.5mA, 12mA and 5 μ A, respectively.

Figure 5 shows the topology of the network we used in our simulators. This is a one-hop network with eight sources and one sink. The sink node is the center node 4 of the nine nodes. Packets from all other nodes are directly delivered to the sink node, but node 0 can't receive any packet from node 2 or node 5. This network with the existing hidden stations and the exposed stations is sufficient to show the basic characteristics of the MAC protocols. We will compare and evaluate the performance of these protocols on the following key metrics: the energy consumption, throughput and latency of the whole network.

3.2 Implementation of the MAC Protocols

The four MAC protocols are implemented as a class hierarchy on top of the physical layer, which is a thin layer encapsulating the RF radio model. The physical layer takes care of low-level synchronization and channel coding. We now briefly discuss the MAC protocols implementation in the experiments and all the parameter setting in the simulator is list in the table 1.

Simplified IEEE 802.11 DCF The simplified IEEE 802.11 DCF [1] (CSMA/CA) protocol without supporting transmission of multiple-fragment was implemented using an 8 byte header encoding the message type (RTS/CTS/DATA/ACK), source and destination ID, sequence number, data length, and CRC. The payload of the DATA packet is 32 bytes. The sequence number is used to detect duplicate packets; retransmissions are triggered upon detection of a missing CTS or ACK packet.

S-MAC The implementation of the S-MAC protocol extends the simplified 802.11 DCF model with a 30% duty cycle. Every period of S-MAC lasts for 1s with an active period of 300ms and a sleep period of 700ms. The control packets and the DATA packet length are set as the same as the 802.11.

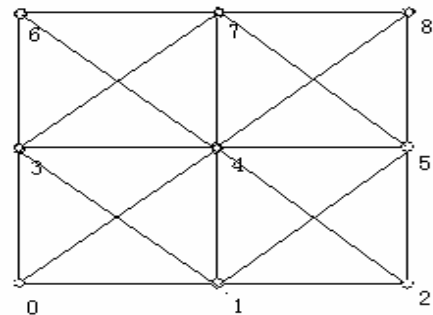


Figure 5 Topology of the network used in the experiment

T-MAC The implementation of the T-MAC protocol enhances the S-MAC model with a dynamic active part to adjust to the load variation. The time-out value T_A is 18ms, which is set to span the contention period (10ms), RTS (1.4ms), and the radio turnover period (10 μ s). The time-out value caused T-MAC to operate with a 2% duty cycle in an empty network. In a loaded network the duty cycle will increase as the active period is adaptively extended. The options for mitigating the early-sleeping problem are out of our focus in this paper here.

PH-MAC The implementation of the PH-MAC protocol has a synchronization period of 50ms, an allocation period of 150ms, and a TDMA period of 800ms, the TDMA part are divided into 16 timeslots, each timeslot can be used to send 8 packets. So a PH-MAC period is 1s, and at start of every period of PH-MAC, the sink node will send a SYNC packet to synchronize all nodes' clock.

3.3 Simulation Results and Discussions

The simulations are carried out by our developed Simulators for the four MAC modules. In the figures, the simplified IEEE 802.11 DCF is denoted as "IEEE 802.11" for short. We randomly inject packets into the 8 edge nodes. By increasing the packet-injecting rate we are able to compare the performance of the four MAC protocols and find out whether or not it collapses under high loads.

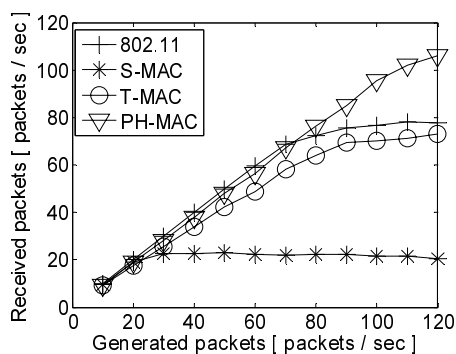


Figure 6 Measured received packets of the sink node

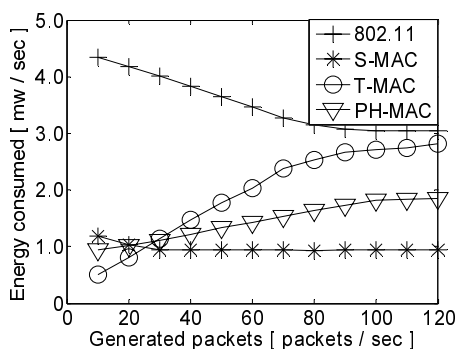


Figure 7 Measured average energy consumption of the network

Average Throughput: Figure 6 plots the received packets versus to the generated packets under the four protocols. As seen from Figure 6, S-MAC performs much poorly because it runs at a 30% duty cycle, this duty cycle is designed to handle 20 packets in every second. On the contrary, T-MAC performs better as it can dynamically adjust to the variable load of the network, and finally it reaches the max throughput that it can handle with. The 802.11 protocol performs very well under light load, but as the load becomes heavy, many nodes are failed due to contention for the channel vehemently, so 802.11 can not deliver more packets. Because PH-MAC is cooperated by TDMA technology, and all the contention work has been done for all the queued packets in the contention period, so the contention in PH-MAC is reduced, and the throughput is thus improved when the load are heavy.

Energy Consumption: Figure 7 shows the energy consumption of each MAC protocol under the homogeneous load. The energy consumption of the 802.11 protocol decreases as the load increases. This is caused by the overhearing avoidance mechanism that shuts down the radio during communication in which a node is not directly involved. The energy consumption of T-MAC initially increases linearly, then reach 2.6 mw under high load, when all nodes are trying their best to deliver packets to the sink. S-MAC is limited by its duty cycle, so it always keeps the minimum energy consumption among all these protocols. As PH-MAC have a contention period, the energy consumed by PH-MAC initially larger than T-MAC, but when the load becomes heavy, the energy consumed by PH-MAC increases slowly as we waste less energy in the TDMA period, while in other protocols the energy are mainly wasted in idle listening, overhearing, and control packets overhead on delivering packets. So although we waste some energy in the contention period to schedule the packet transmission, we can save more energy in the packet transmission, this is the key policy we used to save energy.

Average Latency: Figure 8 shows the one hop latency of each MAC protocol. 802.11 always keep the smallest latency, as it keeps on delivering packets all the time. For

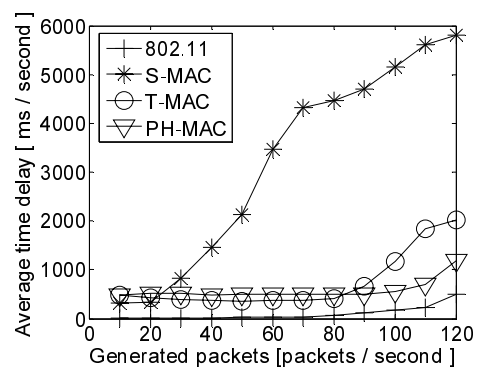


Figure 8 Measured average latency of the network

S-MAC's, the latency increases quickly as the load exceeds the load that S-MAC can handle. T-MAC initially has small latency, when the load becomes larger than it can handle, the latency starts to increase linearly. When the load is light, PH-MAC's latency is little larger than T-MAC, but when the load is larger than it can handle, PH-MAC also starts to increase linearly.

In PH-MAC, packet can only pass through one hop every period, whereas packets in S-MAC or T-MAC can pass through multiple hops in one period. So in order to save more energy, the network latency will become larger in PH-MAC.

4 Conclusion

This paper presents a new MAC protocol for wireless sensor networks termed PH-MAC. By adopting a SYNC-ALLOCATION-TDMA periodical architecture, PH-MAC has a remarkable energy conserving advantage compared to S-MAC and T-MAC. Under this protocol, we are able to make a trade-off between energy and throughput by setting the number of packets that can be sent in a timeslot in the TDMA period; we also can make a trade-off between energy and latency by setting the length of TDMA period of PH-MAC. PH-MAC is particularly suitable to applications of emergency aid. This new protocol is promising for applications in many remarkable advantageous aspects, such as energy consumption, and network throughput. This promise has been verified by simulations.

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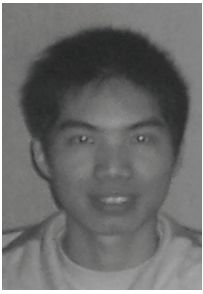
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Biographies



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